

CSC617M: Machine Project Specifications

Interpreter System

I. Overview:

The project is good for a group of three members. The objective is to author and develop a new programming language and implement an *interpreter* software for the new programming language.

II. Minimum Specifications/Requirements:

Statement Constructs:

The following constructs should be present in the new programming language being developed or authored

1. Constant declaration
2. Variable declaration
3. Main program: The main program consists of the following statement constructs:
 - a. Assignment statement
 - b. Input/output statements
 - c. Conditional statement (If – then –else statement)
 - d. Loop statement
 - i. for iteration
 - ii. while loop
 - iii. repeat-until loop
 - e. Functions – formal and actual parameters. You can define your own parameter passing scheme.
- The group will define the sequencing of the declaration portions and the code portion.
- Arithmetic expressions for assignment statement as well as conditional expression (Boolean expressions) for use of the Conditional and Loop statements must be also defined by the authors. Grammar must be capable of handling simple as well as complex expressions.
- Language is capable of handling nested statements.
- Error handling and recovery – system must be able to output and direct users to errors in the code
- *Intermediate Code – the analysis phase generates an intermediate code which would serve as the executable code for interpretation.*

Data Types and Structures:

1. Integers
2. Floating points
3. Strings
4. Characters
5. Arrays
6. *Records*
7. *User-defined Types*
8. **[BONUS] Pointers**
 - *Must be able to demonstrate (e.g. through memory dump, memory debugger window, pointer visualization)*

III. **Programming Language:** The group can implement the interpreter (their own programming language) using any programming language as long as all the members concur.

IV. Submission Details

Milestone/Activites	Deadline
1. Revisit of Compiler Theory Project: - Context-Free Grammar and Lexical rules on proposed programming language - Intermediate Code Specifications - Choice of Programming Language (existing)	Jan. 27, 2026 Draft Version Group Membership
Programming Exercises Milestones: Demonstration of Module - Scanner - Parser	February 10, 2026 March 3, 2026
Interpreter Demo	Marc 17, 2026
Final Project Demo and Submission - Test Cases (sample codes) - Interpreter software binaries - Interpreter source code - User's Manual (on how to use the interpreter and how to program using the programming language) Including Bonus Items: - Language: Pointers - IDE - Debugger - Optimizations - Auxiliary Features	March 24, 2026
Presentation of Final Version to Class and Analysis versus Existing Programming Language – Show the features and capabilities of your interpreter software versus an existing PL. Examples: - Parameter Passing Scheme - Error Detection and Prompting - Error Recovery Methods - Type-checking mechanisms - Runtime Memory Management - Exception Handling - Auxiliary Features Bonus - User-interface and Integrated Development Environment Features Bonus Debug Feature (e.g. Watch and Trace Feature) Bonus – Optimizations implemented - At least 3 optimization techniques - Demonstrate the optimization technique (white box Intermediate Code) - Sample codes to show the strengths and capabilities of	

your new Programming Language	
-------------------------------	--